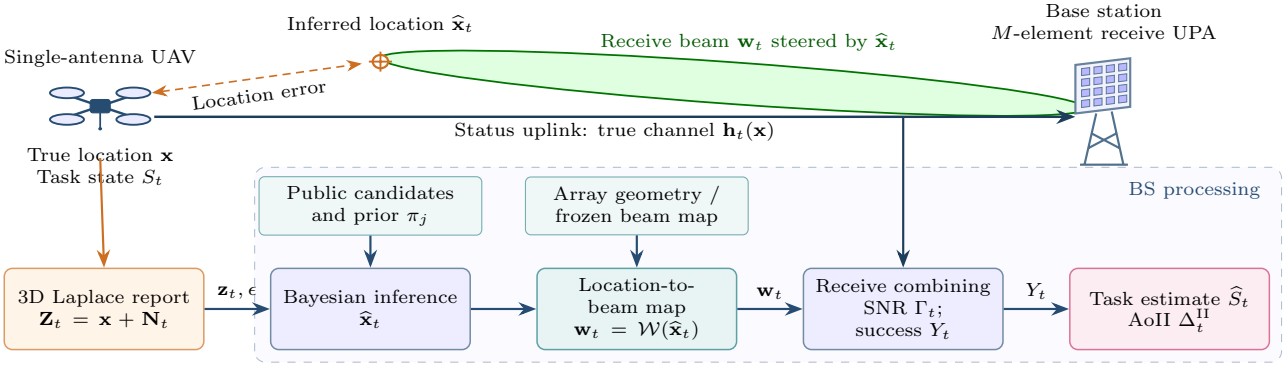

Inferred location $\widehat{\mathbf{x}}_t$
Base station
M-element receive UPA
Single-antenna UAV
Receive beam $\mathbf{w}_t$ steered by $\widehat{\mathbf{x}}_t$
Location error
Status uplink: true channel $\mathbf{h}_t(\mathbf{x})$
True location $\mathbf{x}$
Task state S_t
BS processing
Public candidates and prior π_j
Array geometry / frozen beam map
3D Laplace report $\mathbf{Z}_t = \mathbf{x} + \mathbf{N}_t$
$\mathbf{z}_t, \epsilon$
Bayesian inference $\widehat{\mathbf{x}}_t$
Location-to-beam map $\mathbf{w}_t = \mathcal{W}(\widehat{\mathbf{x}}_t)$
$\mathbf{w}_t$
Receive combining SNR Γ_t; success Y_t
Y_t
Task estimate $\widehat{S}_t$
AoII Δ_t^{II}